Changhyeon Park

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Education

- **KAIST (Korea Advanced Institute of Science and Technology)** *Mar. 2024 - present*
  M.S. in Graduate School of Culture Technology Daejeon, S.Korea
  - Advisor: Prof. Sang Ho Yoon

- **Hongik University** *Mar. 2018 - Feb. 2024*
  B.S. in Computer Engineering Seoul, S.Korea
  - GPA: 4.13/4.5
  - Advisor: Prof. Jaeyoung Park

Research Interests

Sensing & Interaction

I am interested in leveraging diverse sensing technologies to build intuitive interactive systems. While my earlier work centered on haptic feedback, my current research focus has shifted toward **pose estimation using sensing techniques and the interaction systems it enables**. I aim to interpret user context and movement from wearable and mobile sensor data to support more natural and responsive interaction.

Publications

C=Conference, J=Journal, P=Patent, S=In Submission, T=Thesis

[S.1] **Changhyeon Park** and Sang Ho Yoon, Smartwatch multimodal sensing

[C.3] **Changhyeon Park**, Jiwan Kim, Ian Oakley, Cheng Zhang, Sang Ho Yoon UltraHapSense: Simultaneous Ultrasonic Mid-Air Haptics and Hand Localization with Ultrasonic Transducer Array. *ACM SenSys (2027) (Accept Rate : 17.7%, Conditionally Accepted)*

[J.3] **Changhyeon Park**, Yubin Lee, and Sang Ho Yoon. (2025). **UltraBoard: Always-available Wearable Ultrasonic Mid-air Haptic Interface for Responsive and Robust VR Inputs**. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 9, 2, Article 44 (June 2025), 31 pages. https://doi.org/10.1145/3731413

[J.2] **C. Park**, S. Hong and J. Park, (2024). **Effect of Rendering Virtual Vibrotactile Motion on the Perceived Lateral Force**. *IEEE Access*, vol. 12, pp. 173792-173799, doi: 10.1109/ACCESS.2024.3502903.

[J.1] **C. Park**, J. Park, (2024). **Virtual Object Weight Information with Multi-modal Sensory Feedback during Remote Manipulation**. *Journal of Internet Computing and Services*, 25(1), 9–15. https://doi.org/10.7472/JKSII.2024.25.1.9

[C.2] **C. Park***, Y. Sung*, S. Yoon, (2024).**VRmoji: Natural Avatar Movement based on Real-time Facial Expression Recognition System**. *Korea Computer Congress*, 1468-1470.

[C.1] **C. Park**, N. Yoon, J. Park, (2022).**A Multi-Finger Haptic Interface Rendering Resistive Force Using Apparent Tactile Motion**. *Korean Society of Mechanical Engineers*, 2805-2807.

Experience

- **Human-Centered Computing & Intelligent Sening (HiLab), UCLA** *July.2026 – present*
  Visiting Researcher Los Angeles, CA
- **Human-Centered Interactive Technologies Lab (HCITech Lab), KAIST** *Feb.2024 – present*
  Research Assistant Daejeon, Korea
- **Human-Centered Interactive Technologies Lab (HCITech Lab), KAIST** *June.2023 – Sep.2023*
  Research Intern Daejeon, Korea
- **Human-System Interaction Research (HSIR) Lab, Hongik University** *March.2022 – May.2023*
  Undergraduate Research Intern Seoul, Korea

Honors and Awards

- **Academic excellence scholarships, Hongik University** *Spring 2019, Fall 2021, Spring/Fall 2022, Spring 2023*

- **Full-tuition Government Scholarship for Science and Engineering, KAIST** *2024 - present*

Academic Services

- ***Reviewer*, CHI** 2025
- ***Reviewer*, CHI LBW** 2023, 24
- ***Reviewer*, AHs** 2024

Teaching

- ***Teaching Assistant*, GCT623 Interaction Sensing Principle & Application, KAIST** *Spring 2025*